

$$\vec{v}^* = 0, \vec{\omega}^* = 0 \text{ (tako telo)}$$

$$\downarrow \quad \downarrow$$

$$\sum \vec{F}_{zun} = 0 \quad \sum \vec{M}_{zun} = 0$$

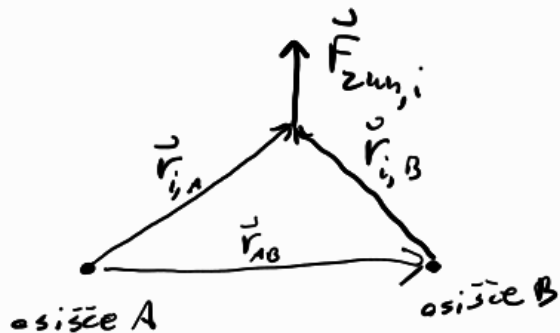
potrebna pogoja za ravnovesje

II

$$\vec{G}^* = 0, \vec{r} = \text{konst.}$$

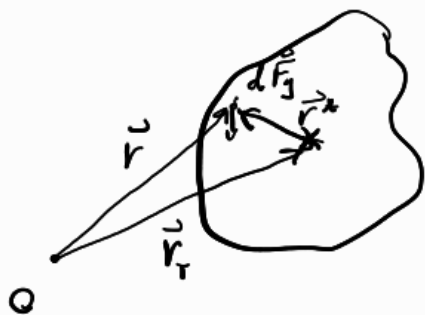
($\vec{G} = \text{konst.}$)

$$\vec{F}_{zun,i} \Rightarrow \vec{M}_{zun,i} = \vec{r}_i \times \vec{F}_{zun,i}$$



$$\begin{aligned} \vec{M}_{zun,A} &= \sum_i \vec{r}_{i,A} \times \vec{F}_i = \\ &= \sum_i (\vec{r}_{i,B} + \vec{r}_{AB}) \times \vec{F}_i = \\ &= \vec{M}_{zun,B} + \underbrace{\vec{r}_{AB} \times \sum_i \vec{F}_i}_{\text{v ravnovesju je enako } 0} \end{aligned}$$

Navar teže:



$$\vec{M}_g = \int dm \vec{r} \times \vec{g} =$$

$$= \int dm (\vec{r}_T + \vec{r}^*) \times \vec{g} =$$

oklepaj velja, če $g = \text{konst.}$ po celotnem telesu

$$= \vec{r}_T \times \int dm \vec{g} + \underbrace{\left(\int dm \vec{r}^* \right) \times \vec{g}}_{=0} =$$

$$= \underline{\underline{\vec{r}_T \times m \vec{g}}}$$

V ravnovesju je navar neodvisen od izbire osišča

Nedoločeni sistem - sistem, ki ga ne moremo fizikalno izračunati
(imamo več neznanj kot enačb)

↓
lahko izračunamo, če dopustimo deformacije

Ravnovesje

- stabilna lega - ob majhnem premiku se vrne v težišče
- labilna lega - ob majhnem premiku se odmakne daleč od težišča
- indiferentna



Deformacije

nateg



$\frac{F}{s}$ - napetost

natezna napetost

$$\vec{F} \parallel \vec{s}$$

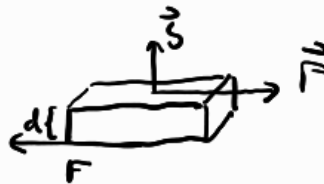
$$\frac{F_x}{s_x}$$

$$\left[\frac{N}{m^2} = Pa \right]$$

deformacija = raztezek (Δl)

napetost = $\underbrace{\text{modul}}_{\text{konst.}} \cdot \text{relativna deformacija}$

strij



strižna napetost

$$\vec{F} \perp \vec{s}$$

$$\frac{F_x}{s_y}$$

deformacija = premik ene ploskve glede na drugo (Δy)

$$\frac{F}{S} = E \cdot \frac{\Delta l}{l}$$

$$\frac{F}{S} = G \frac{\Delta y}{d}$$

E ... prožnostni modul $[\frac{N}{m^2}]$

G ... strižni modul $[\frac{N}{m^2}]$

Tlačna obremenitev:

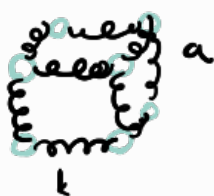
$$\Delta p = -K \frac{\Delta V}{V}$$

K ... stisljivostni modul

$$\frac{\Delta V}{V} = -\chi \Delta p$$

χ ... stisljivost

Hooke $F = -kx$ "vzeti" med atomi snovi v ravnovesju



$$V = Na^3$$

$$a = \sqrt[3]{\frac{V}{N}}$$

$$\nu = \frac{m}{M} = \frac{N}{N_A}$$

št. molov

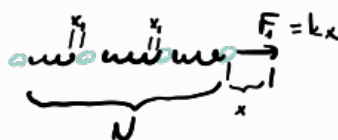
$$\frac{\rho V}{M} = \frac{N}{N_A} \rightarrow \frac{N}{V} = \frac{\rho N_A}{M}$$

Primer: Cu

$$\rho = 8,9 \cdot 10^3 \frac{kg}{m^3}$$

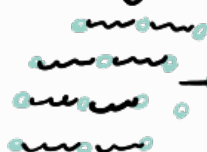
$$M = 63,5 \text{ kg}$$

$$a = \sqrt[3]{\frac{63,5 \text{ kg} \cdot m^3}{8,9 \cdot 10^3 \text{ kg} \cdot 6 \cdot 10^{24}}} \approx 2,5 \cdot 10^{-10} \text{ m}$$



$$x = N_x x_1$$

$$F_1 = k N_x x_1$$



$$N_s \cdot N_x = N$$

$$\frac{F}{S} = \frac{F}{N_s a^2} = E \frac{N_x x_1}{N_x a}$$

$$F = \frac{N}{N_x} k x_1$$

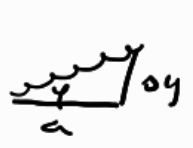
$$E \frac{x_1}{a} = \frac{N k x_1}{N_s a^2 N_x}$$

$$k = E a \quad \text{za baker: } k_1 = E a \approx 20 \frac{N}{m}$$

Pasovno št.

$$\frac{\Delta y}{y} = \mu \frac{\Delta x}{x}$$

v prčni smeri se
snov pri raztezanju
skrči



$$F_y = F \sin \varphi = k a \frac{\varphi^2}{2} \quad \text{za majhne odklone raztezanja}$$

$$\Delta l = \frac{a}{\cos \varphi} - a = a \frac{1 - \cos \varphi}{\cos \varphi} = a \frac{\varphi^2}{2}$$



$$k_1, k_2 \rightarrow E, \mu, G$$

te 3 količine
so povezane

$$G = \frac{E}{2(1+\mu)}$$

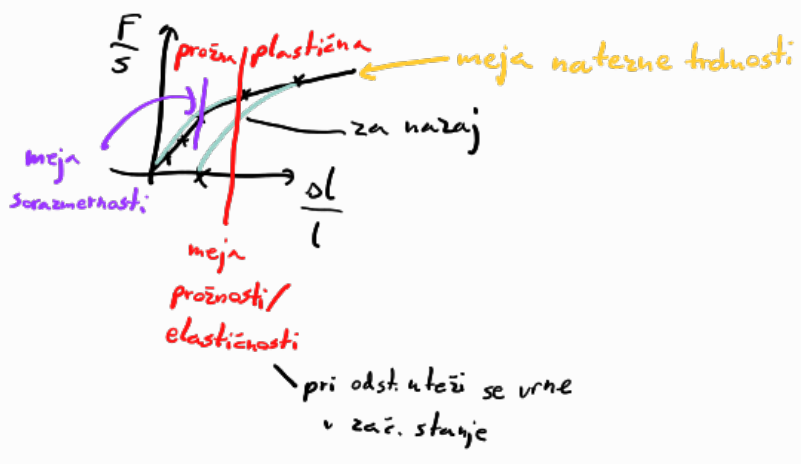
$$\chi \rightarrow E, \mu$$

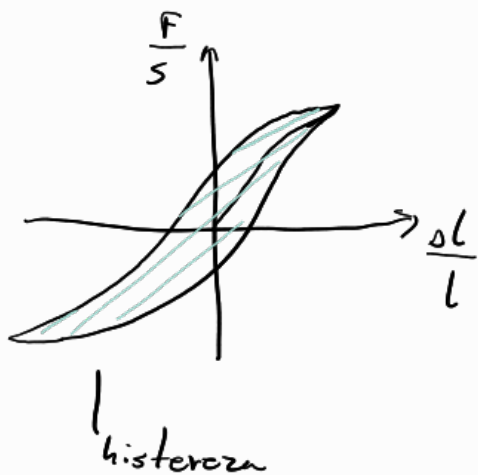


$$\frac{\Delta V}{V_0} = \frac{V - V_0}{V_0} = \frac{(l(1 + \frac{\Delta l}{l}))^3 - l^3}{l^3} = (1 + \frac{\Delta l}{l})^3 - 1 \approx 3 \frac{\Delta l}{l}$$

$$\frac{dV}{V} = d \ln V \approx d \ln l^3 = 3 d \ln l = 3 \frac{dl}{l}$$

$$\chi = \frac{3(1-2\mu)}{E}$$





$$S_{\text{hist.}} = \oint \frac{F}{S} \cdot \frac{dx}{l} = \oint \frac{F dx}{V}$$

\(\downarrow\) delo na enoto volumna